

CMOS VLSI Design Project

Design and Analysis of a CMOS Inverter

Using Cadence Virtuoso

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1 Introduction

The CMOS inverter is one of the most fundamental building blocks of digital integrated circuits. It performs the logical NOT operation, where the output is the complement of the input signal.

In this project, a CMOS inverter was designed and analyzed using Cadence Virtuoso. The complete design flow included schematic creation, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

The Boolean expression of an inverter is

$$Y = \overline{A} \tag{1}$$

where A represents the input signal and Y represents the output.

2 Objectives

The main objectives of this project are:

- To design a CMOS inverter using Cadence Virtuoso.
- To create the transistor-level schematic of the inverter.
- To generate a reusable symbol for the designed inverter.
- To design the physical layout of the CMOS inverter.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation.
- To determine the average power consumption of the inverter.
- To determine the propagation delay of the inverter.

3 Theory of CMOS Inverter

A CMOS inverter consists of one PMOS transistor and one NMOS transistor.

The PMOS transistor forms the pull-up network and is connected between the power supply V_{DD} and the output node. The NMOS transistor forms the pull-down network and is connected between the output node and ground.

The gates of both transistors are connected together to form the input, while their drain terminals are connected together to form the output.

The Boolean expression is

$$Y = \overline{A}. \quad (2)$$

The truth table of the CMOS inverter is shown in Table 1.

Table 1: Truth Table of CMOS Inverter

Input	Output
A	Y
0	1
1	0

When the input is LOW, the PMOS transistor turns ON and the NMOS transistor turns OFF. Therefore, the output is connected to V_{DD} and becomes HIGH.

When the input is HIGH, the PMOS transistor turns OFF and the NMOS transistor turns ON. The output is therefore connected to ground and becomes LOW.

4 Transistor Sizing

The CMOS inverter was designed using one PMOS transistor and one NMOS transistor. From the designed schematic, the approximate transistor dimensions are:

Table 2: Transistor Dimensions Used in the CMOS Inverter

Transistor	Width	Length
PMOS	480 nm	100 nm
NMOS	240 nm	100 nm

Therefore,

$$\frac{W_p}{W_n} = \frac{480}{240} = 2. \quad (3)$$

The PMOS transistor is designed with approximately twice the width of the NMOS transistor. This larger PMOS width helps compensate for the lower mobility of holes

compared with electrons and provides more balanced pull-up and pull-down characteristics.

5 Schematic Design

The CMOS inverter schematic was created using Cadence Virtuoso Schematic Editor.

The circuit contains a PMOS transistor connected between V_{DD} and the output node, while an NMOS transistor is connected between the output node and ground.

The gates of both transistors are connected to the input terminal *in*, and the common drain node forms the output terminal *out*.

The schematic is shown in Figure 1.

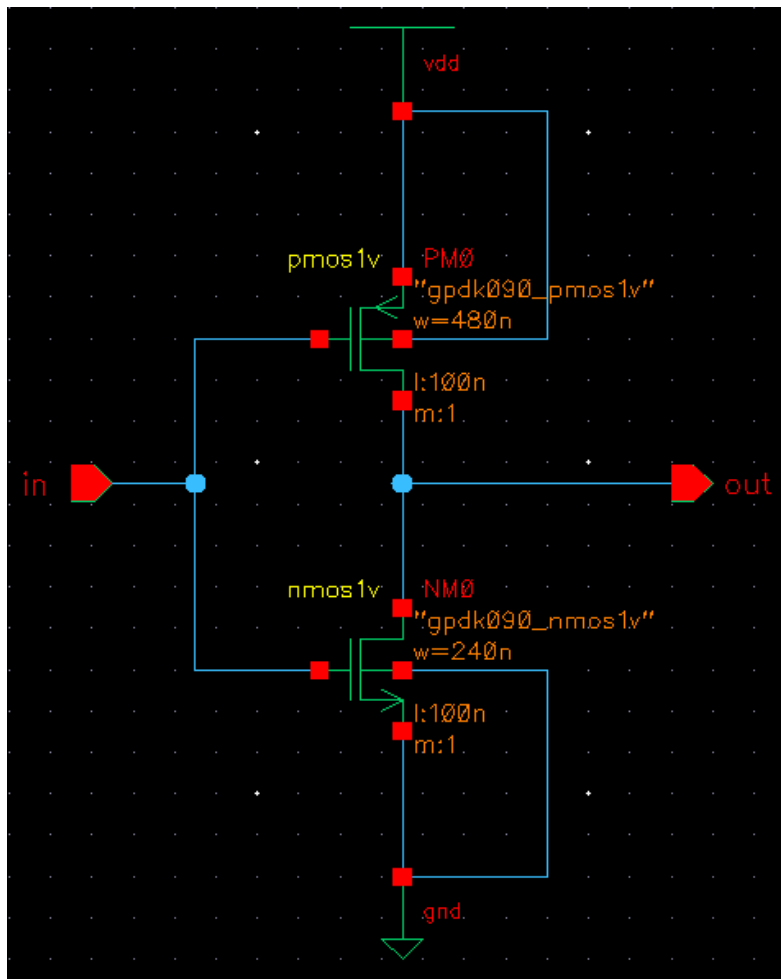


Figure 1: Schematic of the CMOS inverter in Cadence Virtuoso.

The operation of the circuit can be represented as

$$\text{out} = \overline{\text{in}}. \quad (4)$$

6 Inverter Symbol

After completing the schematic, a symbol was generated for the CMOS inverter.

The symbol contains one input terminal labeled *in* and one output terminal labeled *out*.

The generated symbol is shown in Figure 2.

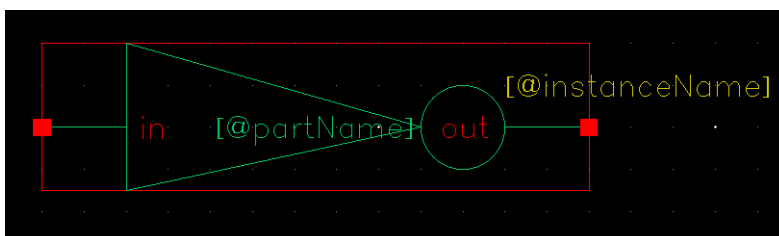


Figure 2: Generated symbol of the CMOS inverter.

The symbol allows the designed inverter to be used conveniently as a hierarchical building block in larger digital circuits.

7 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS inverter was developed using Cadence Virtuoso Layout Editor.

The PMOS transistor was placed in the upper portion of the layout, while the NMOS transistor was placed in the lower portion.

The physical layout contains the required layers for transistor formation and interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- power connection,
- ground connection, and
- input and output routing.

The input signal is connected to the gates of both transistors. The drains of the PMOS and NMOS transistors are connected together to form the output.

The completed inverter layout is shown in Figure 3.

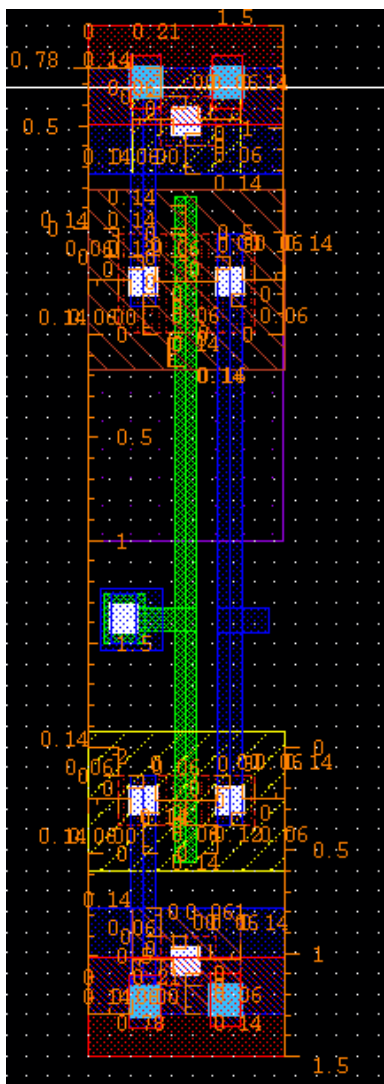


Figure 3: Physical layout of the CMOS inverter.

8 Design Rule Check (DRC)

Design Rule Check is performed to verify that the physical layout satisfies the geometrical fabrication rules of the selected CMOS technology.

DRC checks different parameters such as:

- minimum metal width,
- minimum metal spacing,
- polysilicon width and spacing,
- diffusion spacing,
- contact dimensions,
- contact enclosure,

- well spacing, and
- other fabrication constraints.

The final CMOS inverter layout was checked using the Cadence DRC verification flow.

DRC Result: PASSED

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical design of the inverter is valid with respect to the required fabrication rules.

9 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic verification was performed.

LVS compares the netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connections,
- source and drain connections,
- input terminal,
- output terminal,
- power connection, and
- ground connection.

The final layout successfully matched the schematic.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

This confirms that no unintended connectivity error was introduced during physical layout design.

10 RCX and Parasitic Extraction

After successful DRC and LVS verification, RC parasitic extraction was performed.

Physical interconnections introduce parasitic resistance and capacitance that are not completely represented in an ideal schematic simulation.

These parasitic components influence important parameters such as propagation delay and power consumption.

RCX was therefore performed to extract the parasitic resistance and capacitance of the physical inverter layout.

The extracted view is shown in Figure 4.

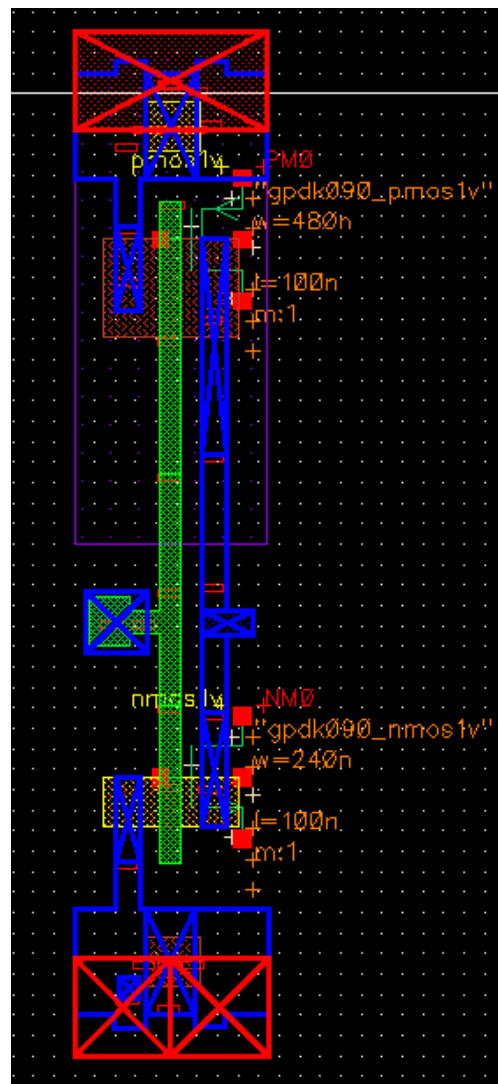


Figure 4: RCX/extracted view of the CMOS inverter.

RCX: Successfully Completed

The extracted inverter circuit was used for post-layout simulation.

11 Post-Layout Simulation

Post-layout transient simulation was performed using the RC-extracted view of the inverter.

Unlike an ideal schematic-level simulation, post-layout simulation includes the parasitic resistance and capacitance introduced by the physical implementation.

Therefore, the extracted simulation provides a more realistic estimation of the circuit performance.

The following parameters were evaluated:

1. Propagation delay
2. Average power consumption

12 Propagation Delay Calculation

Propagation delay represents the time required for a change at the input of the inverter to cause the corresponding change at the output.

Because the inverter produces the complement of the input, an input falling transition generates an output rising transition, while an input rising transition generates an output falling transition.

The two propagation-delay components are commonly represented as

$$t_{pLH} = t_{\text{output rising}} - t_{\text{input falling}} \quad (5)$$

and

$$t_{pHL} = t_{\text{output falling}} - t_{\text{input rising}}. \quad (6)$$

The Cadence Virtuoso Calculator was used to determine the delay between the selected input and output transitions.

A voltage level of approximately 0.6 V was used as the switching reference in the delay calculation.

The delay calculation is shown in Figure 5.



Figure 5: Propagation-delay calculation of the extracted CMOS inverter using the Cadence Virtuoso Calculator.

The measured delay obtained from the Cadence Calculator is

$$t_p = 113.9 \times 10^{-12} \text{ s.} \quad (7)$$

Since

$$1 \text{ ps} = 10^{-12} \text{ s,} \quad (8)$$

the measured propagation delay is

$$t_p = 113.9 \text{ ps} \quad (9)$$

Therefore, the post-layout propagation delay for the selected input-output transition of the CMOS inverter is approximately

$$113.9 \text{ ps}.$$

13 Average Power Calculation

The average power consumption of the inverter was calculated from the post-layout transient simulation using the Cadence Virtuoso Calculator.

The average power over a simulation interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (10)$$

where $p(t)$ represents instantaneous power.

The average of the simulated power waveform was calculated using the Cadence Calculator.

The power calculation is shown in Figure 6.



Figure 6: Average power calculation of the extracted CMOS inverter using the Cadence Virtuoso Calculator.

The obtained average power is approximately

$$P_{\text{avg}} = 2.947 \times 10^{-6} \text{ W.} \quad (11)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (12)$$

the average power becomes

$$P_{\text{avg}} = 2.947 \mu\text{W} \quad (13)$$

Therefore, the measured post-layout average power consumption of the CMOS inverter is approximately

$$2.947 \mu\text{W}.$$

14 Verification and Performance Summary

The overall results of the CMOS inverter design are summarized in Table 3.

Table 3: Summary of CMOS Inverter Design Results

Parameter	Result
Logic Function	$Y = \overline{A}$
PMOS Width	480 nm
NMOS Width	240 nm
Channel Length	100 nm
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	2.947 μ W
Propagation Delay	113.9 ps

15 Discussion

The CMOS inverter was successfully designed and analyzed using the complete custom VLSI design flow in Cadence Virtuoso.

The transistor-level circuit consists of a PMOS transistor acting as the pull-up device and an NMOS transistor acting as the pull-down device.

The PMOS width was chosen as approximately 480 nm, while the NMOS width was approximately 240 nm. Thus, the PMOS width is approximately twice the NMOS width. This sizing helps compensate for the difference between electron and hole carrier mobility.

After schematic verification, a reusable inverter symbol was created. The physical layout was then developed while maintaining the required fabrication rules.

The layout successfully passed DRC, indicating that the physical dimensions and spacing satisfy the technology design rules. The successful LVS result confirms that the layout is electrically equivalent to the intended schematic.

RCX was then performed to extract parasitic resistance and capacitance. The extracted circuit was used for post-layout simulation so that the effect of the physical layout could be included in the performance analysis.

The measured post-layout propagation delay was approximately

$$t_p = 113.9 \text{ ps.} \quad (14)$$

The average power consumption was approximately

$$P_{\text{avg}} = 2.947 \text{ } \mu\text{W.} \quad (15)$$

These values represent the performance of the inverter under the applied simulation conditions after including layout-dependent parasitic effects.

16 Conclusion

A CMOS inverter was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The circuit performs the Boolean operation

$$Y = \overline{A}. \quad (16)$$

The project covered the complete custom digital VLSI design flow, including schematic design, symbol generation, physical layout, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout successfully passed DRC and LVS verification, confirming that the layout satisfies the required fabrication rules and electrically matches the original schematic.

RCX was also completed successfully, allowing parasitic resistance and capacitance to be considered during final simulation.

The measured post-layout performance parameters were

$$\boxed{P_{\text{avg}} = 2.947 \text{ } \mu\text{W}} \quad (17)$$

and

$$\boxed{t_p = 113.9 \text{ ps.}} \quad (18)$$

Therefore, the project successfully demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a CMOS inverter using Cadence Virtuoso.

17 Software Used

The following software tools were used in this project:

- Cadence Virtuoso Schematic Editor

- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator